

CRAB CAKES

Colorado River Area Biologists Computer Aided Knowledge Entry System v2



PIT Crew: Luke Flaker, Mia Gallo, Annaliese Dedmore

Mentor: Smirthya Iyer

Sponsor Agencies: Arizona Game & Fish, USGS, US Fish & Wildlife Service

Project Focus

Improve software used during Colorado River Fish surveys so data is easier to collect, manage, and use.



Why it Matters

The Colorado River is a large support system for people and wildlife. Fish surveys help track changes in population and river health, serving as a conservation guide.

Our Plan

Work with our sponsors

- ❖ Meet with David Rogowski about once a month
- ❖ Stay in touch with the other agencies

Test existing systems

- ❖ Learn more about poor Bluetooth scanner behavior
- ❖ Get more familiar with SHOALS, and the earlier CRAB CAKES version

Look into improvements

- ❖ Linux support and GUI improvements
- ❖ Other wildlife survey programs to better understand survey workflow

Make fish survey data easier to collect, access, and understand!

Current Workflow

Catch → Scan → Measure → Record → Back Up

What happens

- Biologists catch fish using survey methods
- Fish are scanned for tags, measured, and recorded

What's not working

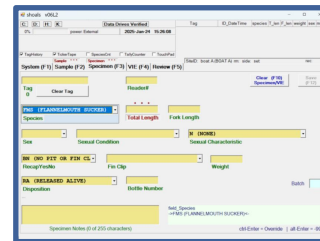
- Bluetooth scanner connection issues
- Previous CRAB CAKES is unreliable
- Difficult interface and no way to graph data

What We Want to Build

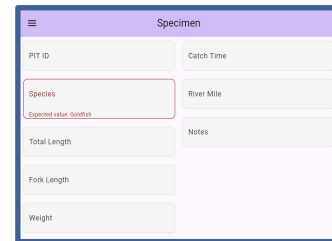
Create a modern, reliable version of CRAB CAKES that is easier for biologists to use in the field.

Main Improvements

- Fix Bluetooth scanner connectivity
- Make the GUI easier to use
- Add Linux support
- Add graphs to visualize fish data and trends
- Make records more user friendly



Shoals legacy system



Previous capstone GUI